

# CILE MD Practice Analysis Summary

Systems: L1P1, L1P2, L2P1, L3P1, L1P3

Trajectory length: 50 ps practice runs. Values are for workflow validation, not final production conclusions.

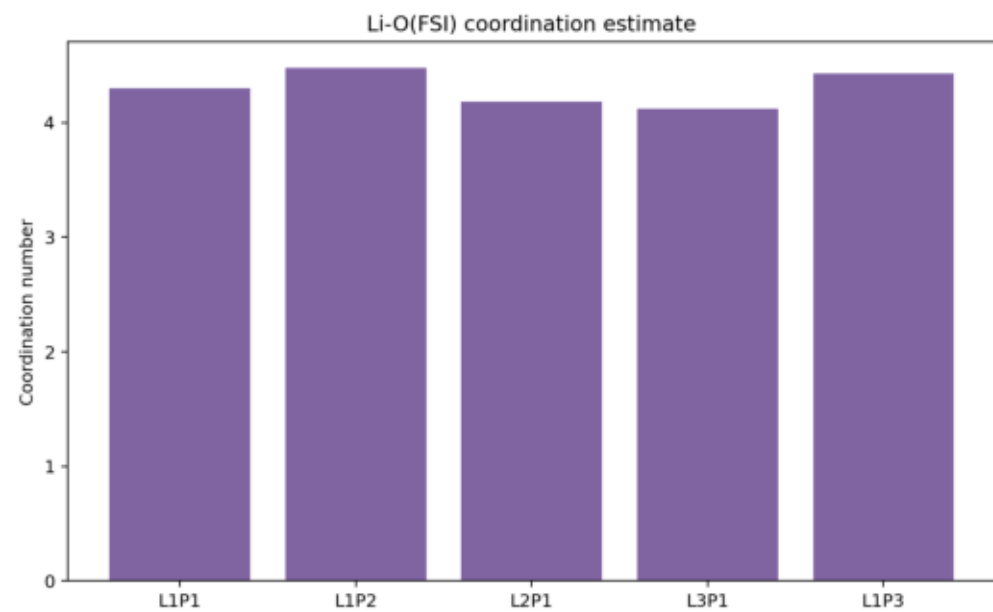
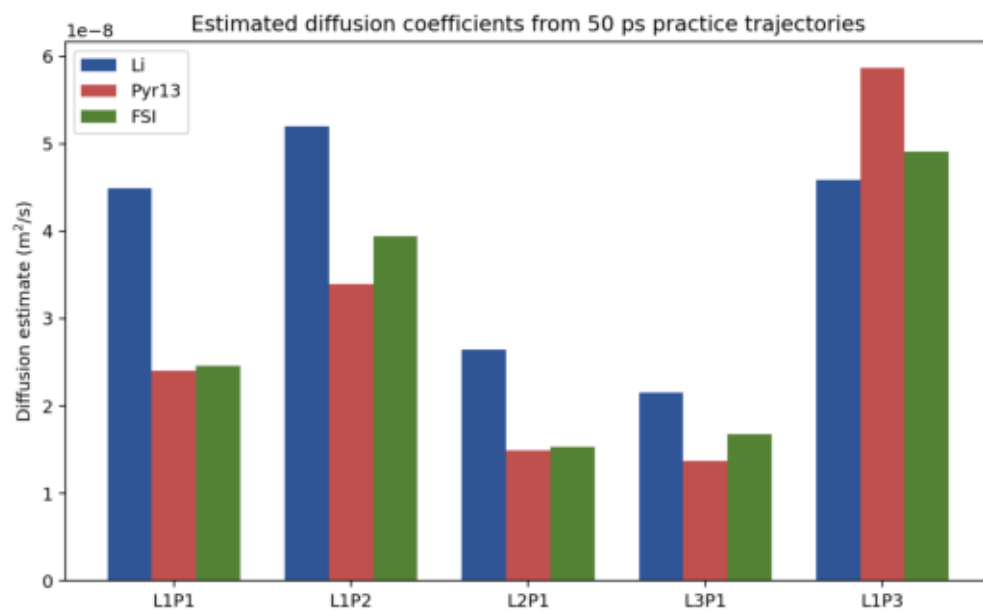
| System | T avg (K) | CN Li-O | D Li     | D Pyr13  | D FSI    |
|--------|-----------|---------|----------|----------|----------|
| L1P1   | 299.0     | 4.30    | 4.49e-08 | 2.40e-08 | 2.46e-08 |
| L1P2   | 299.9     | 4.48    | 5.20e-08 | 3.39e-08 | 3.94e-08 |
| L2P1   | 300.4     | 4.18    | 2.64e-08 | 1.50e-08 | 1.53e-08 |
| L3P1   | 301.8     | 4.12    | 2.15e-08 | 1.37e-08 | 1.68e-08 |
| L1P3   | 298.8     | 4.43    | 4.59e-08 | 5.87e-08 | 4.91e-08 |

Analysis generated:

- Energy / temperature / pressure checks
- Li-O(FSI) RDF and coordination number
- MSD and diffusion estimates from 10-45 ps linear fits

Recommended next step for final research: longer NPT production MD, ideally ns to tens of ns.

# Overall comparison plots



# Temperature stability check

